

# Introduction to the foundations of science and the scientific method

BIOL 1414 (Intro to Cell & Molecular Biology).

**\*\*Main points covered:\*\***

- Biology is the scientific study of life (“bio” = life, “ology” = study of).
  - Ecology is the study of organisms and their environment (“eco” = house).
  - Science is a systematic way of observing the natural world to discover facts and build principles, theories, and laws. It advances by addressing what we *\*don’t\** know, not by simply collecting facts. Science is iterative and circular.
  - Key distinction: Science is a culture of doubt (Feynman); it never proves anything absolute—it can only support or falsify hypotheses.
  - The scientific method cycle: Observation → Question → Hypothesis → Experiment (with data collection) → Evaluation → Sharing results → New hypotheses.
  - A good hypothesis must be **\*\*testable\*\*** and **\*\*falsifiable\*\***.
  - Experimental design terms: control, independent variable (what you change), dependent variable (what you measure), and controlled variables (kept constant).
  - Two types of data: qualitative (descriptions) and quantitative (numbers/measurements).
  - Inductive reasoning (specific observations → general conclusions) vs. deductive reasoning (general principles → specific predictions).
- Hypotheses that survive repeated testing can become theories.  
Theories are well-supported explanations;  
laws describe patterns but do not explain them.  
Science sometimes advances through serendipity (e.g., Fleming’s discovery of penicillin), but observation and follow-up testing are still essential.  
Class activity illustrated sampling bias: a classroom survey on “city vs. rural” hometowns does not represent the world.

Science works by proposing testable ideas, rigorously trying to disprove them, and refining understanding over time through repeated testing and open sharing of results.

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## ### 2. Key Definitions – Concise Study Sheet

| Term   Definition |
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| <b>**Biology**</b>   The scientific study of living things. |
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| <b>**Ecology**</b>   The scientific study of organisms and their environment. |
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| <b>**Science**</b>   The systematic observation of natural events in order to discover facts and formulate principles, theories, and laws. |
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| **Hypothesis** | A proposed, testable, and falsifiable explanation for a phenomenon (an educated guess that can be tested). |  
| **Observation** | Recognizing and noting a fact or occurrence, often involving measurement. |  
| **Data** | Values of qualitative or quantitative variables belonging to a set of items. |  
| **Qualitative data** | Descriptive information that can be observed but not measured (color, texture, smell, appearance, etc.). |  
| **Quantitative data** | Numerical information that can be measured (length, weight, temperature, time, number of individuals, etc.). |  
| **Independent variable** | The factor that is deliberately changed or tested in an experiment ("what I change"). |  
| **Dependent variable** | The factor that is measured or recorded; it depends on the independent variable. |  
| **Control (group)** | The group that receives every element of the experiment except the one being tested. |  
| **Controlled variables** | Factors kept constant so they do not influence the results. |  
| **Falsifiable** | Able to be disproven by evidence. A hypothesis must be falsifiable to be scientific. |  
| **Theory** | A well-supported explanation that summarizes one or more hypotheses that have been repeatedly tested and supported. Stronger than a single hypothesis. |  
| **Law** | A generalization based on repeated observations that describes a pattern in nature but does not explain why it occurs. |  
| **Inductive reasoning** | Moving from specific observations to a broader conclusion (small → big). |  
| **Deductive reasoning** | Using a general principle or theory to predict specific results (big → small). |

#### **Important reminders**

- Science never proves anything absolutely; it can only support or reject hypotheses.
- A theory in science is *not* "just a guess."
- Good experimental design requires clear independent/dependent variables and proper controls.

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### **### 3. Flashcards**

**Front:** What is a hypothesis?

**Back:** A proposed explanation for a phenomenon that must be testable and falsifiable.

**Front:** What is the difference between qualitative and quantitative data?

**Back:** Qualitative = descriptions (color, texture, smell). Quantitative = numbers that can be measured (length, weight, temperature).

**Front:** Independent variable vs. Dependent variable

**\*\*Back:\*\*** Independent = what the experimenter changes. Dependent = what is measured (it depends on the independent variable).

**\*\*Front:\*\*** Why must a hypothesis be falsifiable?

**\*\*Back:\*\*** If it cannot be disproven, it is not scientific. Science advances by trying to disprove ideas.

**\*\*Front:\*\*** Does science prove things?

**\*\*Back:\*\*** No. Science can only support or reject hypotheses. It never provides absolute proof.

**\*\*Front:\*\*** What is a scientific theory?

**\*\*Back:\*\*** A well-supported explanation based on repeated testing of hypotheses. It is much stronger than a single hypothesis.

**\*\*Front:\*\*** What is a scientific law?

**\*\*Back:\*\*** A description of a consistent pattern in nature. It describes *\*what\** happens but does not explain *\*why\**.

**\*\*Front:\*\*** Inductive vs. Deductive reasoning

**\*\*Back:\*\*** Inductive = specific observations → general conclusion. Deductive = general principle → specific prediction.

**\*\*Front:\*\*** What is the role of a control group?

**\*\*Back:\*\*** It receives every element of the experiment except the factor being tested, allowing comparison.

**\*\*Front:\*\*** Example of serendipity in science

**\*\*Back:\*\*** Alexander Fleming's accidental discovery of penicillin when mold contaminated a bacterial culture.

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#### ### 4. Practice Quiz Questions

##### **\*\*Multiple Choice\*\***

1. Which of the following is required for a hypothesis to be scientific?

- a) It must be proven true
- b) It must be testable and falsifiable
- c) It must be based on a theory
- d) It must use only quantitative data

2. In an experiment testing the effect of fertilizer amount on plant growth, the plant height is the:

- a) Independent variable

- b) Dependent variable
- c) Control variable
- d) Hypothesis

3. Science progresses primarily by:
- a) Accumulating more and more facts
  - b) Addressing what we do not yet know
  - c) Proving hypotheses beyond doubt
  - d) Relying only on serendipity

4. A scientific theory is best described as:
- a) An educated guess
  - b) A single experiment's result
  - c) A well-supported explanation based on repeated testing
  - d) A description of a natural pattern with no exceptions

5. Which type of reasoning moves from specific observations to a general conclusion?
- a) Deductive
  - b) Inductive
  - c) Experimental
  - d) Theoretical

**\*\*Short Answer\*\***

6. Explain the difference between a scientific theory and a scientific law.
7. Why is the statement "There is an invisible, heatless dragon living in my office" not a scientific hypothesis?
8. List the main steps of the scientific method in order (as presented in the lecture).

**\*\*Application-Style\*\***

9. You notice that students who sit in the front of the classroom tend to get higher grades.
- Write a testable hypothesis.
  - Identify a possible independent variable and dependent variable.
  - Describe one controlled variable you would keep constant.
  - Explain one reason why a single classroom survey might not be representative.
10. A researcher claims that "all sheep in Scotland are black" after seeing one black sheep. Using the "parable of the black sheep," explain why this claim is not good science and how a more careful scientist would phrase the conclusion.

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### ### 5. Study Guide / Outline (Following Slide Flow)

#### \*\*I. Opening Context\*\*

- Blue Dragon sea slug (\**Glaucus atlanticus*\*) as a real-life example of interesting biology.
- Brief review of previous class (instructor background, Blackboard, expectations).

#### \*\*II. Putting the “ology” in Biology\*\*

- Biology = study of life
- Ecology = study of organisms and their environment (“house”)

#### \*\*III. What is Science?\*\*

- Systematic observation → facts → principles, theories, and laws
- Science advances by investigating the unknown (it is circular/iterative)
- Feynman quote: science is a culture of doubt

#### \*\*IV. Parable of the Black Sheep\*\*

- Illustrates careful, limited conclusions vs. overgeneralization
- Young scientist → middle-career → experienced scientist (increasing caution)

#### \*\*V. Scientific Method Terms\*\*

- Hypothesis (must be testable)
- Observation
- Data

#### \*\*VI. Types of Data\*\*

- Qualitative vs. Quantitative (with examples)

#### \*\*VII. The Scientific Method Cycle\*\*

- Observation → Question → Hypothesis → Experimentation → Evaluation → Sharing results → New hypotheses
- Good hypotheses are falsifiable and testable
- Science never proves anything; it can only disprove or support

#### \*\*VIII. Experimental Design\*\*

- Control group
- Independent variable (what you change)
- Dependent variable (what you measure)
- Controlled variables (kept constant)

#### \*\*IX. Theory, Principle, and Law\*\*

- Hypothesis → (repeated support) → Theory
- Law = describes patterns; Theory = explains them
- Brief notes on Special and General Relativity as examples of theories

## **\*\*X. Types of Logical Thinking\*\***

- Inductive reasoning (specific → general)
- Deductive reasoning (general → specific)
- Examples: brain studies vs. climate/animal distribution

## **\*\*XI. Serendipity in Science\*\***

- Penicillin discovery by Fleming
- Importance of careful observation even in accidents

## **\*\*XII. Class Experiment Example\*\***

- Testing whether ~50% of people live in cities using a classroom survey
- Results (77% city / 23% rural in this sample)
- Discussion of sampling bias and the need for repeated testing in different populations

## **\*\*XIII. Reporting Scientific Work\*\***

- Importance of sharing results (posters, presentations, peer-reviewed papers)
- Peer review adds validity

## **\*\*XIV. Looking Ahead\*\***

- Next lecture continues the study of life

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In replying to your request, I made a screen shot with Google Lens (I Love Google Lens!)

In capturing an image of how the text appears in [Grok.com](https://grok.com).

Google lens *also* provides “AI Overview” captured here:

**Hypothesis, theory, and law** are distinct scientific terms that describe different aspects of how we understand the natural world.

Here is a direct comparison of the three concepts:

| Concept           | Definition                                                   | Purpose                                     | Can it change?                                            | Example                                                        |
|-------------------|--------------------------------------------------------------|---------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------|
| <b>Hypothesis</b> | A proposed, testable explanation for a specific observation. | To guide investigation and experimentation. | Yes, it is easily modified or rejected based on new data. | <i>If a plant receives more sunlight, it will grow taller.</i> |

|               |                                                                                                                |                                                        |                                                                       |                                                                               |
|---------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Theory</b> | A well-substantiated explanation of <i>why</i> or <i>how</i> a natural phenomenon occurs.                      | To explain the underlying mechanisms of nature.        | Yes, it can be refined or updated as new evidence emerges.            | <i>The Theory of Evolution</i> or <i>Germ Theory of Disease</i> .             |
| <b>Law</b>    | A descriptive statement or mathematical equation predicting <i>what</i> will happen under specific conditions. | To summarize a repeating, universal pattern in nature. | Rarely, but it can be restricted or adjusted if exceptions are found. | <i>Newton's Law of Universal Gravitation</i> ( $F = G \frac{m_1 m_2}{r^2}$ ). |

## Key Clarification

A common misconception is that a hypothesis grows into a theory, which eventually matures into a law. This is incorrect. **Theories do not become laws.**

- **Laws** describe *what* happens (patterns).
- **Theories** explain *why* or *how* it happens (mechanisms).

NOte: I am told 22% of AI content is bogus, this remains unverified.